Unit

13 Corrosion of metals and their protection

Pre-study
A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

- The necessary conditions for rusting
- Factors that speed up the rusting process
- How to prevent rusting
- Corrosion resistance of aluminium and aluminium anodization

13.1 What is corrosion? N1

✓ The deterioration of a metal due to its reaction with oxygen, moisture or other substances in the environment is called **corrosion**.

When a metal corrodes, it loses its shine. If the corrosion is serious, the metal may break. Fig. 13.1 shows a corroded gate.

Unreactive metals generally corrode slowly. For example, copper N2 corrodes slowly. Gold is so unreactive that it does not corrode at all (Fig. 13.2).

N1 Do NOT confuse the two words: 'corrosion' and 'corrosive'.

N2 A statue containing copper may have green patches on the surface due to corrosion.



Fig. 13.1 A corroded gate



Fig. 13.2 This giant Buddha is covered with gold in leaf form so it does not corrode

N3 The term 'rusting' only refers to the reaction of iron with air and water. Use the term 'corrosion' for other metals.

13.2 Corrosion of iron: rusting

The corrosion of iron is called **rusting**. **Rust** is the reddish-brown crust N3 that forms on the surface of the iron. Its chemical name is hydrated iron(III) oxide ($Fe_2O_3 \bullet xH_2O$). Rust flakes off the surface of the iron easily. A fresh iron surface is then exposed and so the rusting goes on until the whole piece of iron is disintegrated.

In hydrated iron(III) oxide, the number of water molecules which combine with one formula unit of Fe_2O_3 varies. Therefore x is used to represent the number of water molecules.

corrosion 腐蝕　　rusting 生銹　　rust 銹

13.1
Investigating the necessary conditions for rusting.

Video
Investigating the necessary conditions for rusting

Game
Metal protector

Rusting is a slow process. It usually takes several days for rusting to occur.

Conditions necessary for rusting to occur

We can use the experimental set-up shown in Fig. 13.3 to investigate the conditions necessary for rusting to occur. The iron nails are left in three different environments.

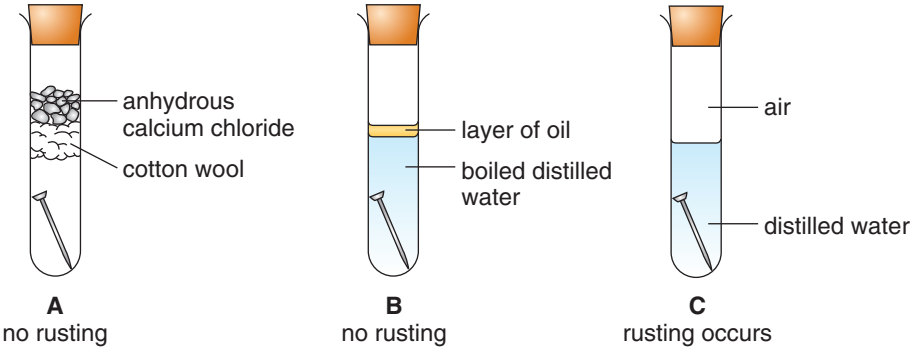


Fig. 13.3 Experimental set-up to investigate the conditions necessary for rusting to occur

Table 13.1 summarizes the observations after one week.

Table 13.1

Observations of iron nail in different environments after one week

Test tube	Environment	Substances present	Has rusting occurred?
A	Anhydrous calcium chloride is a drying agent and absorbs moisture in the air inside the test tube.	Iron and air (oxygen)	No
B	The distilled water is boiled to drive away any oxygen dissolved in it. The layer of oil on the water surface stops oxygen from dissolving in the water again.	Iron and water	No
C	The distilled water contains dissolved oxygen.	Iron, air (oxygen) and water	Yes

The results show that:

✓ For rusting to occur, two things must be present: air (oxygen) and water. N4

N4 When iron is completely immersed in an non-aqueous solvent (e.g. hexane), it does not rust.

What happens during rusting?

During rusting, iron atoms first lose electrons to form iron(II) ions.

$$Fe(s) \longrightarrow Fe^{2+}(aq) + 2e^-$$ CE 2009 Paper 2 Q20

The electrons then combine with oxygen and water to form hydroxide ions.

$$\frac{1}{2}O_2(g) + H_2O(l) + 2e^- \longrightarrow 2OH^-(aq)$$ CE 2009 Paper 2 Q20

The iron(II) ions and hydroxide ions precipitate as iron(II) hydroxide. Iron(II) hydroxide then reacts with dissolved oxygen to form rust, which is hydrated iron(III) oxide ($Fe_2O_3 \bullet xH_2O$).

We can represent the overall change by the following equation:

$$4Fe(s) + 3O_2(g) + 2xH_2O(l) \longrightarrow \underset{\text{rust}}{2Fe_2O_3 \bullet xH_2O(s)}$$

N5 Alkaline solution (for example, NaOH solution) will not speed up rusting. Actually they will inhibit rusting by shifting the position of equilibrium of the following reaction that happens during rusting to the left.

$\frac{1}{2}O_2(g) + H_2O(l) + 2e^- \rightleftharpoons 2OH^-(aq)$

13.2
Investigating factors that influence the speed of rusting of iron.

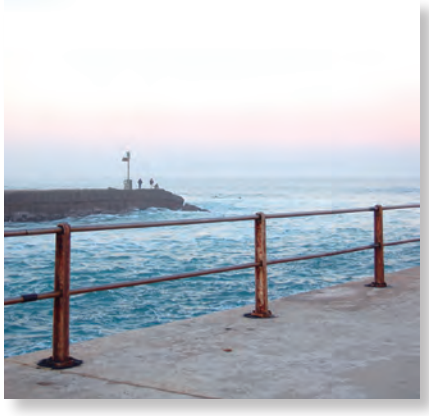


Fig. 13.4 A seaside environment speeds up the rusting process of iron and steel

13.3 What factors can speed up the rusting process?

A number of factors can speed up the rusting process.

1 **Presence of ionic substances** The rusting process speeds up if the water present contains ionic substances such as sodium chloride. Hence a seaside environment speeds up the rusting process (Fig. 13.4) since the thin water layer on the iron surface contains dissolved sodium chloride.

2 **Presence of acidic pollutants** An industrial environment also speeds up the rusting process. This is due to the emission of acidic gases from factories. These gases form acids with moisture in the air, thus speeding up the rusting process. N5

CE 2011 Paper 2 Q42



Fig. 13.5 The exhaust pipe of a car rusts quickly

3 **Higher temperature** At a higher temperature, chemical reaction becomes faster. The rusting process becomes faster too. This is why the exhaust pipes of cars rust quickly (Fig. 13.5). N6

4 **Scratching or bending the surface** Rusting occurs more quickly where the iron surface is scratched or bent.

5 **Attachment to a less reactive metal** The rusting process speeds up if the iron is in contact with a less reactive metal, such as tin and copper. A food can is usually made of mild steel (an iron alloy) coated with a thin layer of tin. When the tin coating of a can is damaged, the steel rusts more quickly than when there is no coating. N7

CE 2008 Paper 1 Q3

N6 The exhaust pipes rust quickly also because acidic gases pass through them.

N7 It is NOT advisable to buy cans of fruit juice if the cans have scratches.

Practice 13.1

1 Explain the following phenomena:

a) Iron articles in a hot desert rust very slowly.
There is very little water moisture in the air.

b) A ship that sunk to the seabed rusts very slowly.
There is very little oxygen in the water near the seabed.

2 The Statue of Liberty is made from copper skin attached to an iron frame.



Statue of Liberty

a) Why does the iron frame rust more quickly when attached to the copper skin?
Copper is less reactive than iron. Iron corrodes instead of copper.

b) The statue sits on an island surrounded by sea water. Why does the sea water increase the rate of rusting?
Sea water contains sodium chloride.

c) Explain why green patches appear on the statue.
Copper(II) ions are formed when the copper corrodes.

13.4 To observe rusting using a rust indicator

We can observe rusting using a **rust indicator**, a mixture of potassium hexacyanoferrate(III) ($K_3Fe(CN)_6$) and phenolphthalein. Potassium hexacyanoferrate(III) gives a blue colour in the presence of iron(II) ions. Phenolphthalein gives a pink colour in the presence of excess hydroxide ions.

We will discuss phenolphthalein in Topic 4 Acids and Bases.

To observe rusting, we place a nail in a gel containing the indicator for one or two days (Fig. 13.6). Fig. 13.7 shows the results.

Notice that the nail is immersed in the gel before the gel sets.

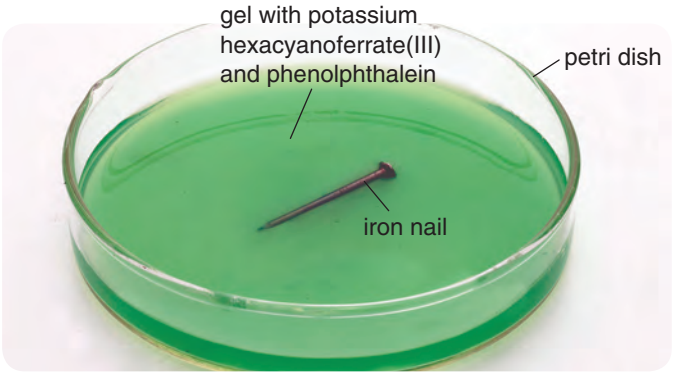


Fig. 13.6 To observe rusting using a rust indicator

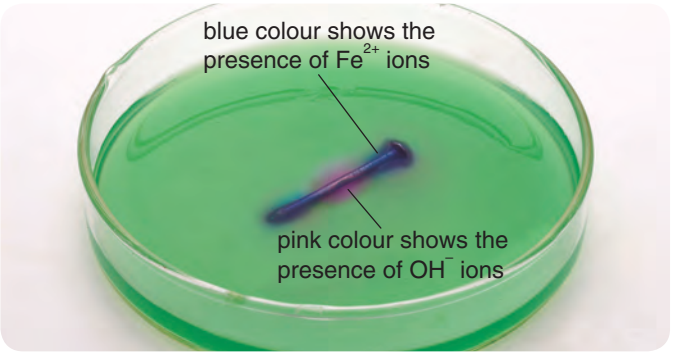


Fig. 13.7 The iron nail rusts. The blue colour shows the presence of Fe^{2+} ions; the pink colour shows the presence of OH^- ions

N8 Paint lasts longer if it is put on a really clean surface. Any trace of water, oil or rust makes the paint peels off more easily.

13.3
Investigating the effectiveness of various ways to prevent rusting.

Video
A method protecting food cans from rusting



Fig. 13.8 Painting on the steel frame of a bridge prevents rusting

13.5 How to prevent rusting?

We are now going to discuss ways to prevent rusting. Most ways involve coating the iron with something to keep out the oxygen and water.

Using protective coatings

Coating iron articles with paint, oil, grease or plastic

Painting is a cheap and simple way of protecting iron from rusting. Cars, N8 ships, bridges and iron railings are all protected by painting (Fig. 13.8). A layer of paint prevents both oxygen and water from reaching the iron beneath. However, as soon as the paint is scratched, the metal is exposed to air, and rusting starts (Fig. 13.9). Therefore painting is not suitable for protecting the moving parts of machines.

CE 2009 Paper 2 Q6
DSE 2013 Paper 1A Q7

rust indicator 鐵銹指示劑

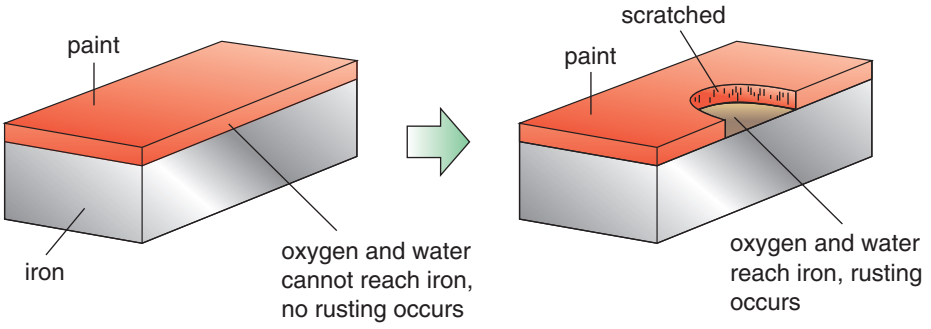


Fig. 13.9 A layer of paint prevents both oxygen and water from reaching the iron beneath



Fig. 13.10 Iron paper clips coated with plastic

Oil or grease can also prevent oxygen and water from reaching the iron. They are used to protect the moving parts of machines which cannot be painted. Oil or grease serves as a lubricant also. However, the oil or grease must be reapplied constantly. Thus, this method is unsuitable for large structures. N9

Coating an iron article with plastic is more expensive than painting (Fig. 13.10).

N9 Bicycle gear wheels are protected by greasing.

Tin-plating

DSE 2012 Paper 1A Q16

A food can is usually made of mild steel coated with a thin layer of tin (Fig. 13.11). This method is called **tin-plating**. The tin protects the steel from both oxygen and water. Furthermore, tin ions are non-poisonous. N10

N10 Examination questions often test whether students know that 'tin CANNOT prevent iron from rusting via sacrificial protection' as tin is less reactive than iron.



Fig. 13.11 Food cans made of mild steel are coated with tin

tin-plating 鍍錫

Zinc-plating

CE 2008 Paper 1 Q3
DSE 2012 Paper 1A Q16

Iron can be coated with a layer of zinc. Iron coated with zinc is called **galvanized iron** and this method is called **zinc-plating** (Fig. 13.12). The zinc layer protects the iron from oxygen and water. When the zinc coating is damaged, zinc 'sacrifices' itself to protect the iron. This is because zinc is more reactive than iron. Thus, zinc corrodes instead of iron. N11

N11 Zinc can protect iron from rusting in two ways:
- For an iron object fully plated with zinc, the zinc layer protects the iron from oxygen and water.
- If part of the zinc layer is scratched to expose the iron underneath, the zinc still protects the iron by sacrificial protection.

Students should be able to distinguish between these two ways.



Fig. 13.12 The external parts of many *squatters* in Lei Yue Mun are made of galvanized iron

CE 2007 Paper 2 Q48
CE 2009 Paper 2 Q6

However, galvanized iron is not suitable for making food cans because zinc ions are poisonous.

Electroplating

We can plate a very thin layer of metal on iron by an electrical process called **electroplating**. Chromium and nickel plating can protect iron from contacting with oxygen and water. This plating is not easily chipped off. Chromium-plated items look more attractive. However, the cost of chromium plating is quite high. Bumpers of cars were once chromium-plated (Fig. 13.13).

CE 2009 Paper 2 Q6

We will further discuss electroplating in Topic 5 Redox Reactions, Chemical Cells and Electrolysis.

Bumpers of cars are usually made of plastic now.



Fig. 13.13 Chromium-plated bumpers can be found on many vintage cars

galvanized iron 鍍鋅鐵　zinc-plating 鍍鋅　squatter 寮屋　electroplating 電鍍

Using sacrificial metals

CE 2007 Paper 2 Q48

When a lump of a more reactive metal is attached to a piece of iron, the more reactive metal corrodes instead of iron. The more reactive metal 'sacrifices' itself to prevent iron from rusting. This method is thus called **sacrificial protection**. N12

N12 Students should be able to give a correct description of sacrificial protection, such as 'attaching / joining magnesium / zinc to the iron object', NOT 'placing magnesium / zinc near to the iron object'.

Zinc and magnesium are two metals which are used in this way. An underground pipeline and a ship's hull is protected by joining a piece of magnesium or zinc to it (Figs. 13.14–13.15). The sacrificial metals must be replaced regularly before they corrode away completely. N13

N13 Zinc and magnesium prevent iron from losing electrons.

CE 2011 Paper 2 Q36

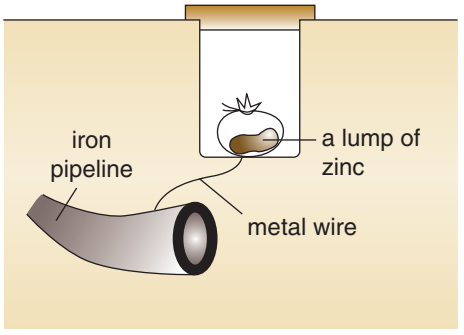


Fig. 13.14 Zinc 'sacrifices' itself to protect the underground iron pipeline from rusting



Fig. 13.15 Pieces of zinc protect the ship's hull from rusting

Impressed current cathodic protection

We can protect iron from rusting by connecting it to the negative terminal of a battery while a conductor such as graphite is connected to the positive terminal◂ (Fig. 13.16). This supplies electrons to the iron and prevents the formation of iron(II) ions. This method is called **impressed current cathodic protection**.

◂ The electrode connected to the negative terminal of the battery is called the cathode. The electrode connected to the positive terminal is called the anode.

A car body is protected from rusting by connecting it to the negative terminal of the car battery.

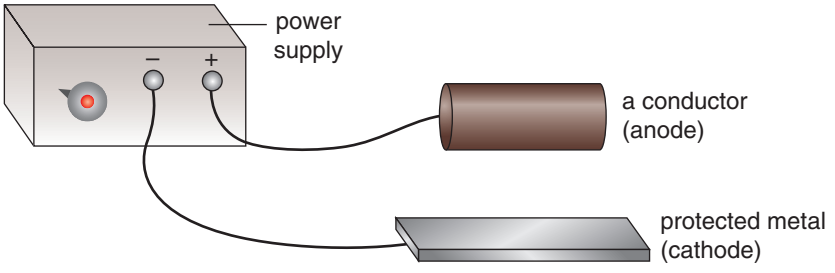


Fig. 13.16 An impressed current cathodic protection system

sacrificial protection 犧牲性保護　　impressed current cathodic protection 外加電流陰極保護



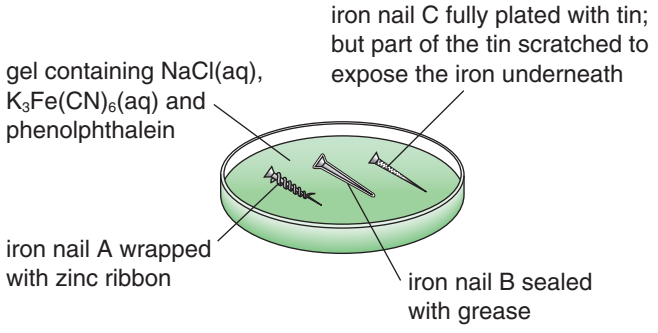
Fig. 13.17 Cutlery made of stainless steel

Using corrosion resisting alloys of iron

Stainless steel is an alloy made up of iron, chromium and nickel. Stainless steel does not corrode easily (Fig. 13.17). Nickel and chromium are much more expensive than iron. Therefore using alloys is the most expensive way to prevent rusting. However, this method is the best because no maintenance is needed.

Example 13.1

Q The diagram below shows an experimental set-up for investigating the factors affecting rusting.



a) What would be observed if an iron nail in the above set-up rusts? Explain your observation.

b) Decide whether each of the iron nails in the above set-up would rust or not during the experiment? Explain your answer in each case.

A

a) Blue and pink colours would be observed near the iron nail that rusts.

When rusting occurs, both iron(II) ions and hydroxide ions are formed.

$K_3Fe(CN)_6$(aq) gives a blue colour in the presence of iron(II) ions. Phenolphthalein gives a pink colour in the presence of excess hydroxide ions.

b) Both iron nails A and B would not rust. Iron nail C would rust.

Iron nail A would not rust because zinc provides sacrificial protection.

Iron nail B would not rust because it is sealed with grease. This prevents oxygen and water from reaching the iron.

Iron nail C would rust because the exposed iron is in contact with oxygen and water / the exposed iron is in contact with tin (a less reactive metal).

CE 2005 Paper 2 Q10 & 11
DSE 2012 Paper 1B Q9

stainless steel 不銹鋼

Practice 13.2

1 Suggest the principle of the rust prevention method used in each of the following cases:

a) galvanized iron buckets
Zinc provides sacrificial protection.

b) greasing the steel gear system of a bicycle
Grease prevents oxygen and water from reaching the iron.

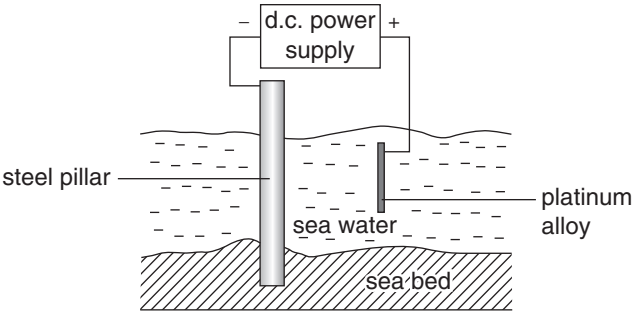
c) tin-plating iron food cans The tin layer prevents oxygen and water from reaching the iron.

d) painting an iron gate
Paint prevents oxygen and water from reaching the iron.

e) attaching magnesium pieces to the hull of a ship
Magnesium provides sacrificial protection.

f) chromium-plated car bumpers
The chromium plating prevents oxygen and water from reaching the iron.

2 The following diagram shows a system used in some piers for slowing down the rusting of steel pillars:



Suggest how the system can prevent steel pillars from rusting. The negative terminal of the d.c. power supply supplies electrons to the steel pillar and prevents the formation of iron(II) ions.

CE 2010 Paper 1 Q9

Video
- Problems brought about by rusting
- Application of rusting

13.6 Socioeconomic implications of rusting

One important use of iron is to make steel. Steel is widely used as a construction material in vehicles, machinery, bridges, pipelines, etc.

In our daily lives, we can easily see rusting on vehicle body panels, outdoor furniture and metal tools. Rusting affects the useful service lives of our possessions.

Rusting may endanger our safety. Rusting of reinforcing steel in concrete may cause structural failures of highways, buildings and bridges (Figs. 13.18–13.19). These can endanger public safety and require significant repair costs.

Several studies over the past 30 years have shown that the annual direct cost of corrosion to an industrial economy is approximately 3.1% of gross national product. In the United States, this amounts to over US$276 billion per year.

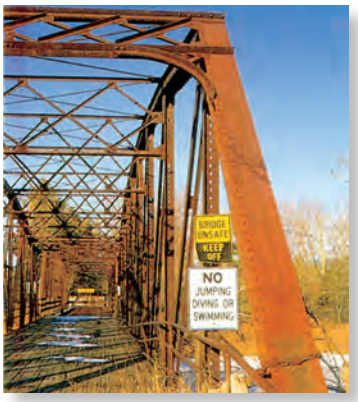


Fig. 13.18 Rusting has made this bridge unsafe



Fig. 13.19 Corroded reinforcing steel in concrete

13.7 What is the most suitable method to protect a particular iron article from rusting?

We have learnt that there are different methods to protect iron from rusting (Table 13.2). The methods are different both in their effectiveness and cost.

Table 13.2

Different methods of protecting iron from rusting			
Method	**Effectiveness of protection**	**Appearance of finished product**	**Relative cost**
Painting	fair	colour of paint	low
Plastic coating	fair	colour of plastic	low
Tin-plating	fairly good	dull silvery	medium
Zinc-plating	good	dull silvery	medium
Chromium-plating	good	bright and shiny	high
Alloying (e.g. using stainless steel)	excellent	bright and shiny	very high

The following are some factors you should consider when choosing the most suitable method to protect an iron item from rusting:

1 How long does the item need to last?

2 Where is the item used?

3 What is the item used for?

4 Is appearance important?

5 Is the cost high?

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Decision Making

For each of the following cases, choose a method to protect the iron or steel article from rusting. Give reason(s) for each choice.

1 To protect a bus stop sign.

2 To protect the base of an electric iron.

3 To protect the steel frame of a swing in a park.

4 To protect the body of an old car.

5 To protect steel scissor blades.

6 To protect a steel pencil-sharpener.

7 To protect steel railings along the pavement.

8 To protect a food can.

Decision Making

There is no single right answer for this type of exercises. It is more important that students are able to support their choice with sensible reason(s) rather than giving the right answer in this type of exercises. The following answers are just for reference.

1 Painting is used because it is cheap.
2 Chromium plating is used because both good protection and shiny appearance are important.
3 Painting is used because it is cheap.
4 Painting is used because it is cheap.
5 Chromium plating is used because both good protection and shiny appearance are important.
6 Zinc plating or tin plating is used because both methods give good protection with medium cost.
7 Painting is used because it is cheap.
8 Tin plating is used because tin is non-poisonous and the method gives fairly good protection with medium cost.

13.8 Corrosion resistance of aluminium

When aluminium reacts with oxygen in the air, an even coating of aluminium oxide forms. This oxide layer sticks firmly to the surface of the metal and is *impermeable* to oxygen and water. It protects the metal beneath from further attack (Fig. 13.20). The oxide layer does not fall off. Rusting of iron is destructive but oxidation of the aluminium surface is a useful process. N14

N14 Advantages and disadvantages of using aluminium over iron:
- advantages
  - aluminium is lighter;
  - aluminium is more corrosion resistant;
  - aluminium can be dyed more easily.
- disadvantages
  - aluminium is more expensive;
  - aluminium is not so strong.

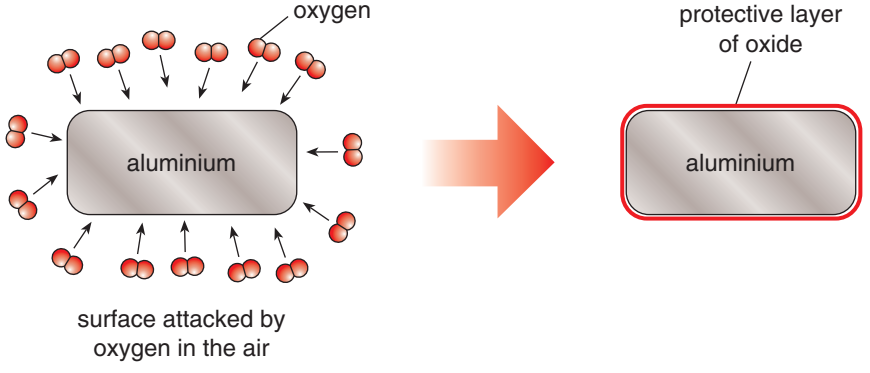


Fig. 13.20 When aluminium reacts with oxygen in the air, the oxide layer formed sticks to the surface of the metal

impermeable 不能滲透的

Aluminium anodization

CE 2005 Paper 2 Q48

Anodization is an electrolytic process used to increase the thickness of the metal oxide layer on the surface of a metal part. The process is called 'anodization' because the metal part to be treated forms the anode of the electric circuit. N15

Notice that anodization CANNOT improve the strength of aluminium.

Aluminium is anodized to increase its corrosion resistance and surface hardness and to allow dyeing more easily. Anodized aluminium is used for making masts (Fig. 13.21) for yachts and *windsurfing boards*, window frames, cooking utensils, milk bottle caps, drink cans (Fig. 13.22) and chocolate wrappings (Fig. 13.23). N16

CE 2006 Paper 2 Q34
CE 2009 Paper 2 Q41
CE 2011 Paper 2 Q46



Fig. 13.21 A mast made of anodized aluminium



Fig. 13.22 Drink cans made of anodized aluminium



Fig. 13.23 Milk bottle caps and chocolate wrappings made of anodized aluminium

N15 Examination questions often ask about the anodization of aluminium:
- how the process is carried out;
- how the process makes aluminium more corrosion resistant.

Anodized aluminium is NOT an alloy. Anodization CANNOT increase the electrical conductivity of aluminium.

N16 Anodized aluminium is more commonly used than iron for making window frames because anodized aluminium is more corrosion resistant than iron. However, anodized aluminium is NOT as hard as iron.

Experimental set-up for aluminium anodization

CE 2009 Paper 2 Q47

Fig. 13.24 shows the experimental set-up for aluminium anodization. The aluminium object to be anodized is connected to the positive terminal of the power source. An aluminium sheet is rolled into cylindrical shape and connected to the negative terminal. The **electrolyte** is dilute sulphuric acid. During the process, oxygen produced at the anode reacts with the aluminium object to increase the thickness of the oxide layer.

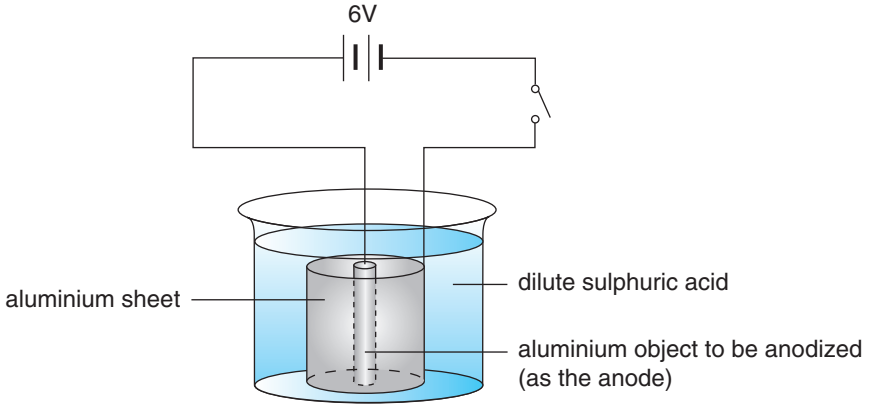


Fig. 13.24 Experimental set-up for aluminium anodization

DSE 2013 Paper 1A Q6

anodization 陽極電鍍　windsurfing board 滑浪風帆板　electrolyte 電解質

Practice 13.3

1 Explain why aluminium does not corrode in moist air.

The aluminium oxide layer sticks firmly to the surface of the metal and stops oxygen / air / water from reaching the metal.

2 Suggest THREE advantages of using anodized aluminium to make drink cans.

The drink cans can be dyed more easily.
The surface hardness of the drink cans can be increased.
The corrosion resistance of the drink cans can be increased.

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

Chemistry Magazine

Stainless steel

Most of the mined iron is used to make steel, an alloy of iron.



Fig. 13.25 A car engine which is made of stainless steel

In 1908, Harry Brearley, a 36-year-old British metallurgist, was given the opportunity to head up a research laboratory in Britain.

Brearley was given the task of finding an alloy which would prolong the life of gun barrels, which were affected by the high temperatures that developed inside the barrels when the guns were fired. Around 1913, he found that a mixture containing 12.8% chromium and 0.24% carbon was extremely resistant to chemical attack. He called his alloy 'rustless steel', but it was decided that 'stainless steel' was a better name.

After Brearley left the firm, his successor W.H. Hatfield continued the research and made the type of stainless steel that we use today.

Brearley's chromium steel formed the basis for the wide range of stainless and special steels which are now used across the world.

Questions

Chromium is in between zinc and iron in the reactivity series.

1 Suggest how chromium can prevent iron in stainless steel from rusting. By sacrificial protection / chromium reacts with oxygen or water more readily than iron.

2 Coating chromium on iron-made objects can prevent the objects from rusting. Explain how this process can prevent rusting. Chromium coating can prevent iron from contacting with oxygen and water.

3 In an experiment, a piece of chromium is put into magnesium sulphate solution.

Predict whether a reaction will take place. Explain your answer. No.
Chromium is less reactive than zinc. Zinc is less reactive than magnesium.
∴ chromium is less reactive than magnesium, so chromium cannot displace magnesium from magnesium sulphate solution.

Have you mastered?

Key terms

corrosion 腐蝕 113 rusting 生銹 113 hydrated iron(III) oxide 水合氧化鐵(III) 113

rust indicator 鐵銹指示劑 117 potassium hexacyanoferrate(III) 六氰合鐵(III)酸鉀 117

phenolphthalein 酚酞 117 tin-plating 鍍錫 118 galvanized iron 鍍鋅鐵 119

zinc-plating 鍍鋅 119 electroplating 電鍍 119 sacrificial protection 犧牲性保護 120

impressed current cathodic protection 外加電流陰極保護 120 stainless steel 不銹鋼 121

aluminium anodization 鋁的陽極電鍍 125 electrolyte 電解質 125

Checklist

After studying this unit, you should be able to

- ☐ describe the nature of iron rust;
- ☐ describe the essential conditions for the rusting of iron;
- ☐ describe and explain factors that influence the speed of rusting of iron;
- ☐ describe the observations when a rust indicator (a mixture of potassium hexacyanoferrate(III) and phenolphthalein) is used in an experiment that investigates rusting of iron;
- ☐ describe and explain the methods of rusting prevention as exemplified by
  - coating with paint, oil, grease or plastic;
  - tin-plating;
  - galvanizing;
  - electroplating;
  - sacrificial protection;
  - impressed current cathodic protection;
  - alloying;

- ☐ be aware of the socioeconomic impact of rusting;
- ☐ understand why aluminium is less reactive and more corrosion resistant than expected;
- ☐ describe how the corrosion resistance of aluminium can be enhanced by anodization.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 The deterioration of a metal due to its reaction with oxygen, moisture or other substances in the environment is called corrosion.

2 The corrosion of iron is called rusting. Rust is the reddish-brown crust that forms on the surface of the iron. Its chemical name is hydrated iron(III) oxide ($Fe_2O_3 \bullet xH_2O$).

3 For rusting to occur, two things must be present: air (oxygen) and water.

4 Factors that speed up the rusting process include:

a) presence of ionic substances;

b) presence of acidic pollutants;

c) higher temperature;

d) scratching or bending the surface; and

e) attachment to a less reactive metal.

5 Rusting can be observed using a rust indicator, a mixture of potassium hexacyanoferrate(III) ($K_3Fe(CN)_6$) and phenolphthalein.



6 Methods to prevent rusting include:

a) coating with paint, oil, grease or plastic;

b) tin-plating;

c) zinc-plating;

d) electroplating;

e) using sacrificial metals;

f) impressed current cathodic protection; and

g) using corrosion resisting alloys of iron.

7 a) When aluminium reacts with oxygen in the air, an even coating of aluminium oxide forms.

b) This oxide layer sticks firmly to the surface of the metal and is impermeable to oxygen and water.

c) The thickness of the oxide coating can be increased by aluminium anodization. This increases the corrosion resistance of aluminium.

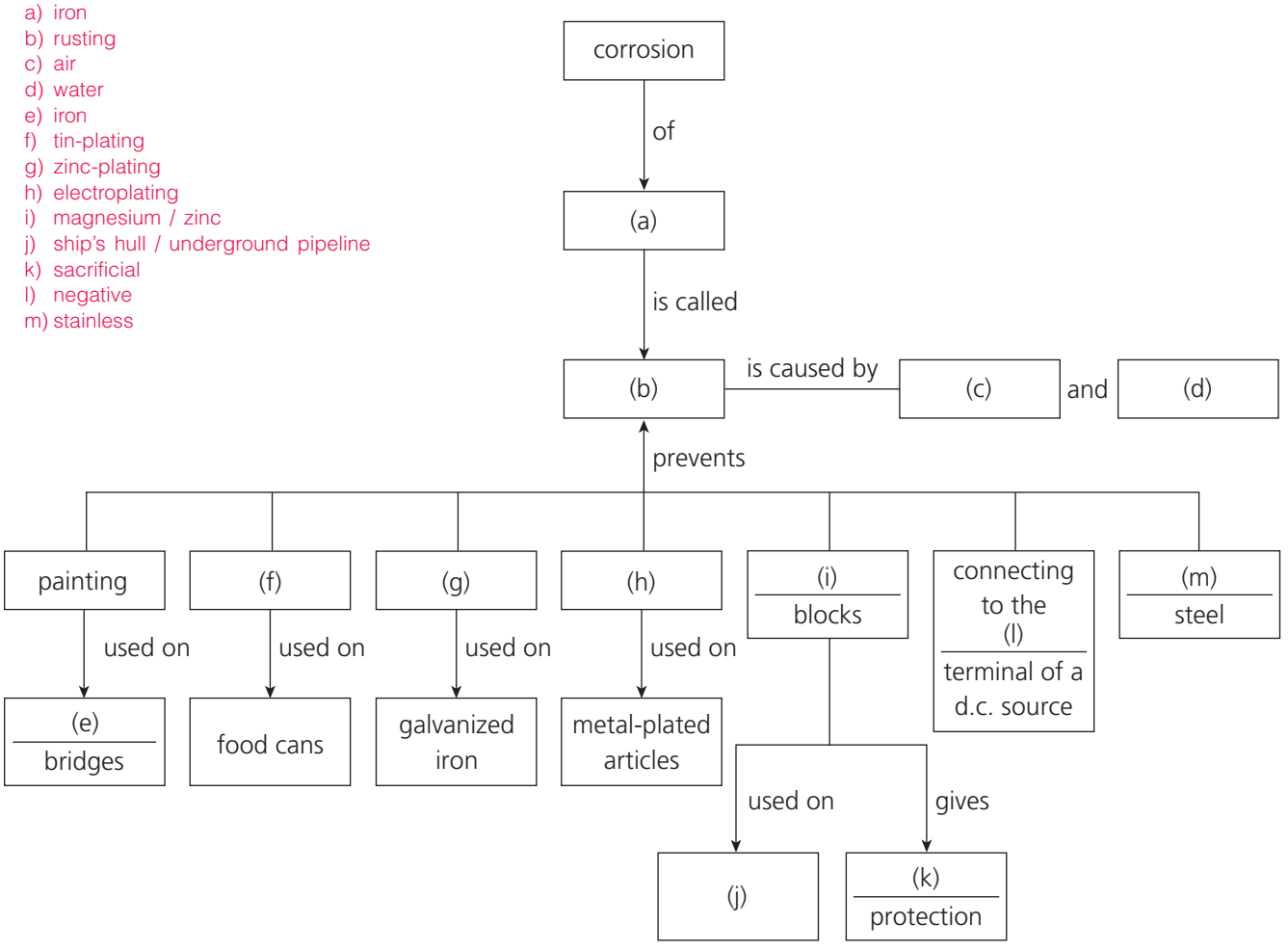
Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the following concept map.

a) iron
b) rusting
c) air
d) water
e) iron
f) tin-plating
g) zinc-plating
h) electroplating
i) magnesium / zinc
j) ship's hull / underground pipeline
k) sacrificial
l) negative
m) stainless



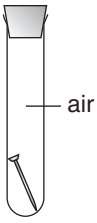
2 List the factors that can speed up the rusting process.

- Presence of ionic substances
- Presence of acidic pollutants
- Higher temperature
- Scratching or bending the surface
- Attachment to a less reactive metal

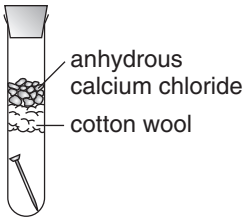
Part II Multiple choice questions

3 The diagrams below represent four tests in an experiment designed to find out the necessary conditions for rusting. Which diagram shows the most favourable conditions for rust to appear on the nail?

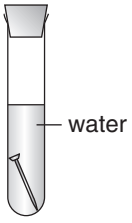
A



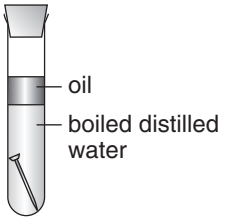
B



(C)



D



Water contains dissolved oxygen. Rusting occurs in the presence of air (oxygen) and water.

Directions: Questions 4 and 5 refer to the following experiment.

The iron nails in the following glass dishes were treated in different ways. A rust indicator solution containing potassium hexacyanoferrate(III) and phenolphthalein was poured into the dishes to cover the nails.



4 What colour would be observed if an iron nail in the glass dish rusts?

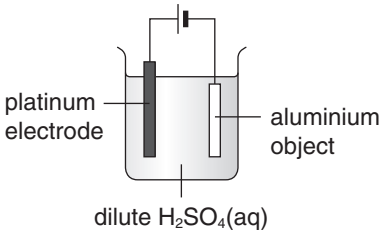
(A) Blue
B Brown
C Red
D Yellow

5 In which of the dishes would the iron nail rust?
Refer to page T52 for explanation.

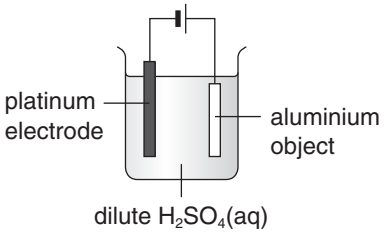
A Dish 1 only
(B) Dish 2 only
C Dish 1 and dish 3 only
D Dish 2 and dish 3 only

6 Which of the set-ups shown below can best be used to anodize an aluminium object?

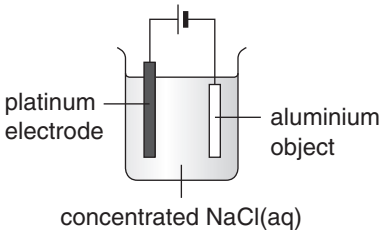
A



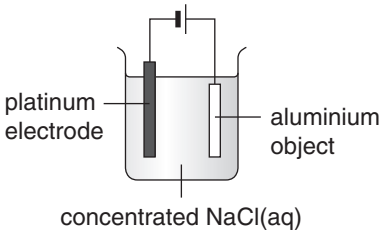
(B)



C



D



(HKDSE, 2013, Paper 1A, 6)

7 In order to prevent rusting, zinc blocks can be attached to the surface of steel ships. This is because

A zinc is a stronger oxidizing agent than iron.
B zinc prevents iron from losing electrons.
C zinc separates iron from air and water.
D zinc removes oxygen from rust.

(HKCEE, 2011, Paper 2, 36)

8 Which of the following rust prevention methods does NOT match with the iron-made object?

	Rust prevention method	Iron-made object
A	painting	gate
B	greasing	machinery parts
C	zinc plating	food can
D	chromium plating	car bumper

(HKCEE, 2009, Paper 2, 6)

9 Which of the following half equations are involved when iron rusts?

(1) $Fe \longrightarrow Fe^{3+} + 3e^-$
(2) $Fe \longrightarrow Fe^{2+} + 2e^-$
(3) $Fe^{2+} \longrightarrow Fe^{3+} + e^-$

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(HKCEE, Paper 2, 2009, 20)

10 Which of the following is / are example(s) of sacrificial protection against rusting?

(1) Iron dustbins coated with tin
(2) Underground iron pipes connected to magnesium bars
(3) Iron railings coated with paint

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(1) Tin is less reactive than iron. It CANNOT provide sacrificial protection.
(3) Paint does NOT provide sacrificial protection.

11 Which of the following changes would occur after an aluminium article has been anodized?

(1) Its tensile strength increases.
(2) It is more easily dyed.
(3) It is more corrosion resistant.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(1) Anodization does NOT increase the tensile strength of an aluminium article.

12 Which of the following substances is / are alloy(s)?

(1) Anodized aluminium
(2) Bronze
(3) Galvanized iron

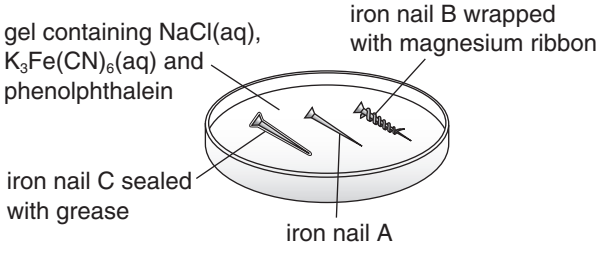
A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(2) Bronze is an alloy of copper.

Part III Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

13 The diagram below shows an experimental set-up for investigating the factors affecting rusting.



a) What would be observed if an iron nail in the above set-up rusts?

b) Suggest which of the iron nails in the above set-up would NOT rust during the experiment. Explain your answer.

(HKDSE, Paper 1B, 2012, 9)

Unit Exercise

14 Four iron-made objects are placed separately in gel with rust indicator solution containing potassium hexacyanoferrate(III) and allowed to stand in air for some time. Complete the following table by writing down the observation and giving the relevant explanation for each of the cases.

Case	Observation	Explanation
Iron-made object fully plated with zinc		
Iron-made object fully plated with tin		
Iron-made object fully plated with zinc, but part of the zinc scratched to expose the iron underneath		
Iron-made object fully plated with tin, but part of the tin scratched to expose the iron underneath		

(HKCEE, Paper 1, 2008, 3)

15 Explain the following:

a) Steel hot water tanks often have a bar of magnesium suspended in the water.

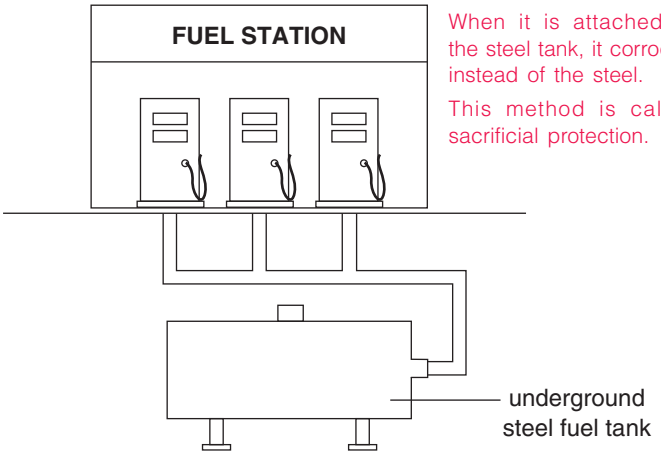
b) A *battered* tin can rusts more quickly than a damaged galvanized iron rubbish bin.

a) Magnesium is a sacrificial metal.
It corrodes instead of the steel tank.

b) Tin is less reactive than iron. Once the tin coating of the can is damaged, the iron rusts more quickly than when it is alone.
Zinc is more reactive than iron. Even when the zinc coating of the iron rubbish bin is damaged, the zinc corrodes instead of the iron.

16 Steel fuel tanks are buried underground at fuel stations.

b) ii) Zinc / magnesium is more reactive than iron.
When it is attached to the steel tank, it corrodes instead of the steel.
This method is called sacrificial protection.



a) Explain why the steel tank in the soil rusts easily.
Both oxygen and water are present in the soil.

b) i) Suggest how the steel tank can be protected from rusting.
Attach a lump of zinc / magnesium to the tank.

ii) Explain the principle of the method suggested in (i).

c) The tank could be made from stainless steel.

i) Suggest a reason for this.
Stainless steel does not corrode easily.

ii) Why is stainless steel NOT used? Stainless steel is very expensive. It is cheaper to protect the steel tank from rusting and replace the tank when necessary.

17 a) Zinc is used to prevent iron rusting. The iron is coated in zinc. This method of protection is called galvanizing. The layer of zinc stops oxygen and water reaching the iron. Galvanizing is an example of sacrificial protection. Zinc is more reactive than iron. Zinc corrodes instead of iron.
Explain how this sacrificial protection works.

b) When iron rusts, the atoms of iron, Fe, are changed into iron(II) ions, Fe^{2+}.

Write an equation for this reaction.
$Fe(s) \longrightarrow Fe^{2+}(aq) + 2e^-$

c) Zinc reacts with copper(II) sulphate solution to make copper. A solution of zinc sulphate is also made.

Write down an equation for this reaction.
$Zn(s) + CuSO_4(aq) \longrightarrow ZnSO_4(aq) + Cu(s)$

(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 2, Jun. 2011, 9)

batter 破爛的

18 John wants to stop his bicycle chain rusting.



a) John puts oil onto his bicycle chain.

Explain why this stops the chain rusting.
The oil prevents oxygen and water from reaching the chain.

b) Iron can have a piece of magnesium attached to it. This also stops iron rusting.

Explain how. Magnesium is more reactive than iron. Magnesium corrodes instead of iron.

c) Write an equation for the rusting of iron.
$4Fe(s) + 3O_2(g) + 2xH_2O(l) \longrightarrow 2Fe_2O_3 \bullet xH_2O(s)$

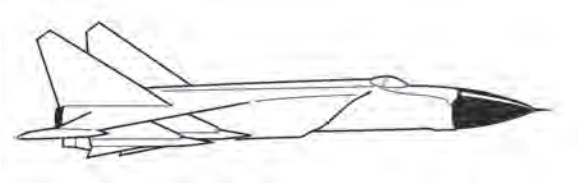
(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 2, Jan. 2012, 14)

19 Metals are a very useful type of material.

The table below shows some information about four metals.

Metal	Melting point (°C)	Density (g cm^{-3})	Relative strength (1 is very weak)	Relative hardness (1 is very soft)
Aluminium	660	2.7	11	2.8
Copper	1 085	8.9	33	3.0
Iron	1 538	7.9	20	4.5
Titanium	1 668	4.5	40	6.0

a)



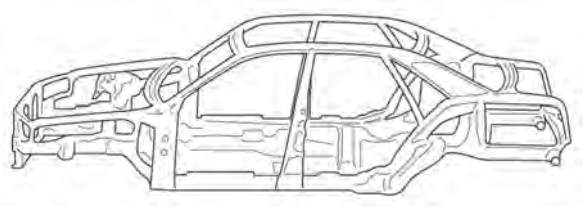
Titanium is used to make this aeroplane. Use information from the table to explain why.
Low density, strongest

b) Aluminium reacts with oxygen in the air to form an even coating of oxide.
This oxide layer sticks firmly to the surface of the metal and stop oxygen / air / water from reaching the metal.

b) Iron corrodes quickly in moist air. This is called rusting. Aluminium does not corrode in moist air.

Explain why.

c) Steel is an alloy that contains mostly iron. The diagram below shows a car body.



Aluminium or steel can be used to make a car body. One advantage of aluminium is that it will not corrode in moist air.

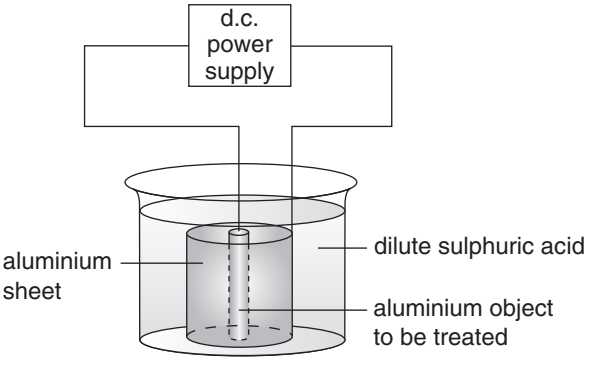
Describe ANOTHER advantage and ONE disadvantage of using aluminium instead of steel to make a car body. Information in the table may help you answer this question. Refer to the next page.

(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 1, Jan. 2008, 7)

20 The surface of aluminium metal is always covered with a thin layer of another substance.

a) Give the chemical name of the thin surface layer and state the effect it has on aluminium.
Refer to the next page.

b) This thin surface layer can be made thicker by treating the aluminium using electrolysis as shown below.



i) Give the name of the process.
Aluminium anodization

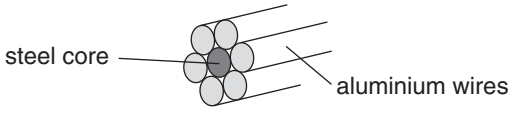
ii) Suggest whether the aluminium object to be treated should be made the positive or negative electrode.
The positive electrode

Topic Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Multiple choice questions

1 The diagram shows the structure of an overhead power cable. The steel core gives strength to the cable.



Aluminium is used because it is a good conductor of electricity and

(A) it has a low density.
B it is a good conductor of heat.
C it can be hammered into shape.
D it is a hard metal.

(AQA GCSE (Foundation & Higher Tier), Chemistry, Unit C1a, Jun. 2011, 8(d))

2 Which of the following statements concerning an aluminium ore consisting mainly of Al_2O_3 is correct?

(Relative atomic masses: O = 16.0, Al = 27.0)

A Carbon can be used to extract aluminium from this ore.
B The abundance of this ore in the earth crust is very low.
C This ore contains more than 55% of aluminium by mass.
(D) Aluminium can be extracted from this ore due to the advancement of technology in applying electricity.

(HKDSE, Paper 1A, 2012, 9)

!!3 Metal X is more reactive than platinum but less reactive than iron. Which of the following statements is most likely to be correct?

A X burns vigorously in air to form an oxide.
B X reacts with dilute hydrochloric acid vigorously.
C X displaces iron from iron(II) sulphate solution.
(D) The oxide of X can be reduced by heating with carbon. The oxide of iron can be reduced by heating with carbon. As X is less reactive than iron, the oxide of X should also be reduced by heating with carbon.

!!4 X and Y are different metals. X reacts with the solution of nitrate of Y according to the following equation:

$$X(s) + Y^{2+}(aq) \longrightarrow X^{2+}(aq) + Y(s)$$

Which of the following deductions is correct?

A Both X and Y can react with water.
B The reactivity of Y is higher than that of X.
(C) Atoms of X lose electrons in the reaction.
D X can be obtained by heating its oxide with Y.

5 In an oxide of metal M, the mass percentage of M is 55.0%. What is the chemical formula of this oxide?

(Relative atomic masses: O = 16.0, M = 39.1)

(A) MO_2
B M_2O
C M_2O_2
D M_2O_3

(HKDSE, Paper 1A, 2012, 3)

19 c) Advantage
Any one of the following:
- Aluminium is less dense so that the car body will be lighter.
- Aluminium is easier to shape / bend.

Disadvantage
Any one of the following:
- Aluminium is weaker than steel.
- Aluminium is more expensive.
- Aluminium is not easily welded.

20 a) Aluminium oxide layer
When aluminium reacts with oxygen in the air, an even coating of aluminium oxide forms.
This oxide layer sticks to the surface of the metal.
It protects the metal underneath from further attack.

4 Option B — X can displace Y from a solution of $Y^{2+}(aq)$ ions. Thus, the reactivity of X is higher than that of Y.
Option C — Atoms of X lose electrons to form $X^{2+}(aq)$ ions in the reaction.

6 An ore contains 80% of zinc sulphide by mass. Assuming that the other components in this ore do not contain zinc, what mass of the ore is required to extract 0.70 g of zinc?

(Relative atomic masses: S = 32.1, Zn = 65.4)

A 0.88 g
B 1.04 g
C 1.30 g
D 1.76 g

(HKCEE, Paper 2, 2011, 8)

7 The following equation represents the reaction of an oxide of iron with carbon:

$$Fe_3O_4(s) + 2C(s) \longrightarrow 3Fe(s) + 2CO_2(g)$$

What mass of the iron oxide is required to produce 22.1 g of iron?

Refer to page T53 for explanation.

(Relative atomic masses: O = 16.0, Fe = 55.8)

A 15.9 g
B 30.5 g
C 91.8 g
D 126 g

8 Ammonium dichromate ($(NH_4)_2Cr_2O_7$) decomposes on heating to give chromium(III) oxide, water and nitrogen. What mass of water is obtained when 278 g of ammonium dichromate undergo complete decomposition?

(Relative atomic masses: H = 1.0, N = 14.0, O = 16.0, Cr = 52.0)

A 19.8 g
B 39.6 g
C 79.2 g
D 158 g

$(NH_4)_2Cr_2O_7(s) \longrightarrow Cr_2O_3(s) + 4H_2O(l) + N_2(g)$
278 g ? g

Number of moles of $(NH_4)_2Cr_2O_7$ = $\frac{\text{mass of } (NH_4)_2Cr_2O_7}{\text{molar mass of } (NH_4)_2Cr_2O_7}$

= $\frac{278 \text{ g}}{252.0 \text{ g mol}^{-1}}$

= 1.10 mol

According to the equation, 1 mole of $(NH_4)_2Cr_2O_7$ decomposes to produce 4 moles of H_2O.

∴ number of moles of H_2O obtained = 4 x 1.10 mol
= 4.40 mol

Mass of H_2O required = number of moles of H_2O x molar mass of H_2O
= 4.40 mol x 18.0 g mol^{-1}
= 79.2 g

∴ 79.2 g of water are obtained.

9 Consider the following equation:

$$X_2(g) + 3Y_2(g) \longrightarrow 2XY_3(g)$$

4 moles of $X_2(g)$ are allowed to react with 6 moles of $Y_2(g)$. What is the number of moles of $XY_3(g)$ formed?

$X_2(g)$ + $3Y_2(g)$ ⟶ $2XY_3(g)$
4 moles 6 moles ? moles

A 4
B 6
C 8
D 10

According to the equation, 1 mole of X_2 reacts with 3 moles of Y_2 to produce 2 moles of XY_3.

2 moles of X_2 react with 6 moles of Y_2 to produce 4 moles of XY_3.

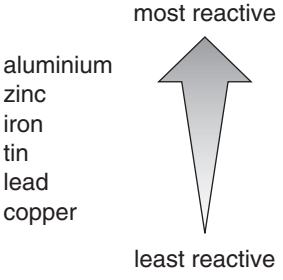
10 When equal mass of Mg and Zn granules is added separately to excess dilute hydrochloric acid, Mg gives more hydrogen than Zn does. Which of the following explanations is correct?

Refer to page T53 for explanation.

A The reactivity of Mg is higher than that of Zn.
B The metallic bond in Mg is stronger than that in Zn.
C The relative atomic mass of Mg is smaller than that of Zn.
D Mg is an alkaline earth metal while Zn is a transition metal.

Directions: Questions 11–12 refer to the following reactivity series.

Reactivity of some metals



11 The metal tin occurs in the Earth's crust as the compound called cassiterite, SnO_2. How would you expect tin to be extracted from its oxide?

A By mixing the oxide with copper and heating strongly
B By electrolysis of the solid oxide
C By mixing the oxide with carbon and heating strongly
D By electrolysis of a solution of the oxide

Tin is less reactive than iron and more reactive than lead. Both iron and lead can be extracted from their oxides by heating with carbon. Thus, tin should be extracted from its oxide in the same way.

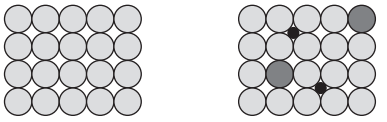
Tin is less reactive than iron. It does not react with substances in the environment.

12 The main reason why food cans made from steel (iron) are coated with tin is because

A tin has a lower density than iron.
B tin is less reactive than iron.
C tin is a better conductor of heat than iron.
D tin is stronger than iron.

(AQA GCSE (Foundation Tier), Chemistry, Unit C1a, Jun. 2012, 6(a)–(b))

Directions: Questions 13–14 refer to the following diagrams that show the arrangement of atoms in pure iron and in stainless steel.



pure iron stainless steel

13 Pure iron is soft and easily shaped because the atoms

A are not arranged in a regular pattern.
B are small and spherical.
C are in layers that can slide over each other.
D are of different sizes.

(AQA GCSE (Foundation & Higher Tier), Chemistry, Unit C1a, Jun. 2011, 8(a))

14 Stainless steel rather than pure iron is used for making cutlery. Which row in the table gives two reasons why stainless steel is used?

Stainless steel

A	is harder.	is a better heat conductor.
B	is more resistant to corrosion.	is harder.
C	is a better heat conductor.	can be more easily shaped.
D	can be more easily shaped.	is more resistant to corrosion.

(AQA GCSE (Foundation & Higher Tier), Chemistry, Unit C1a, Jun. 2011, 8(b))

Stainless steel contains iron atoms and atoms of other elements of different sizes. It is less easily shaped than iron.

15 Both the frame and gear system of a bicycle are made of steel. Which of the following combinations can be used to prevent these parts of the bicycle from rusting?

	Frame	Gear system
A	painting	greasing
B	painting	galvanizing
C	tin-plating	greasing
D	tin-plating	galvanizing

(HKDSE, Paper 1A, 2013, 7)

16 Which of the following are the advantages of using anodized aluminium to make drink cans?

(1) The drink cans can be dyed more easily.
(2) The hardness of the drink cans can be increased.
(3) The corrosion resistance of the drink cans can be enhanced.

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(HKCEE, Paper 2, 2011, 46)

17 Which of the following combinations is / are correct?

Object	Corresponding corrosion prevention method / principle
(1) aluminium window frames	cathodic protection
(2) galvanized iron buckets	sacrificial protection
(3) tin-plated iron cans	alloying

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKDSE, Paper 1A, 2012, 16)

Directions : *Each question (Questions 18–20) consists of two separate statements. Decide whether each of the two statements is true or false; if both are true, then decide whether or not the second statement is a correct explanation of the first statement. Then select one option from A to D according to the following table :*

A Both statements are true and the 2nd statement is a correct explanation of the 1st statement.
B Both statements are true but the 2nd statement is NOT a correct explanation of the 1st statement.
C The 1st statement is false but the 2nd statement is true.
D Both statements are false.

	1st statement	2nd statement	
18	When excess magnesium ribbons are added to iron(II) sulphate solution, the solution gradually changes from pale green to yellow.	When magnesium ribbons are added to iron(II) sulphate solution, a displacement reaction occurs. *(HKCEE, Paper 2, 2011, 30)*	C
19	Galvanized iron is used for making food cans.	Zinc can prevent iron from rusting by sacrificial protection. *(HKCEE, Paper 2, 2007, 48)*	C
20	When iron and copper are separately immersed in hexane completely, iron corrodes faster than copper.	Iron can be oxidized more readily than copper. *(HKDSE, Paper 1A, 2012, 23)*	C

Part II Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

21 a) Iron is extracted from iron ore. Iron ore contains iron oxide. There are different types of iron oxide. Which of these formulae corresponds to the oxide with the highest proportion of iron atoms? Circle the correct answer.

(FeO) Fe_2O_3 Fe_3O_4

b) Iron is extracted from iron oxide in a blast furnace. Iron forms through a sequence of reactions.

Here are the reactions, but they are not in the correct order.

A $2C + O_2 \longrightarrow 2CO$
B $FeO + CO \longrightarrow Fe + CO_2$
C $Fe_3O_4 + CO \longrightarrow 3FeO + CO_2$
D $3Fe_2O_3 + CO \longrightarrow 2Fe_3O_4 + CO_2$

Put the reactions into the correct order. The last one has been done for you.

start | A | D | C | B | end

c) What mass of iron is present in 72.0 g of FeO?

(Relative atomic masses: O = 16.0, Fe = 55.8)

d) Carbon monoxide is a gas at room temperature.

What does this indicate about the structure and bonding of carbon monoxide?

(OCR GCSE 21st Century Science (Higher Tier), Additional Science A, Unit 2, A216/02, Jan. 2010, 1)

22 Both copper and aluminium have many uses. Give a different use for each of these metals and give a property of the metal on which that use depends.

(Edexcel IGCSE (Higher Tier), Chemistry, Paper 2, Nov. 2008, 6(d))

Refer to page T54.

21 c) Mass of iron present = 72.0 g x $\frac{55.8}{55.8 + 16.0}$
= 56.0 g

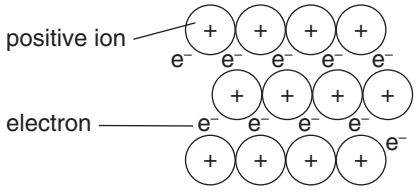
d) Any two of the following:
- small molecules;
- far apart;
- weak forces between the molecules;
- strong forces (covalent bonds) between atoms.

23 b) ii) There are strong attractions between the positive ions and delocalized electrons.
A lot of heat is needed to overcome the strong attractions.

23 a) Copper is used to make electric wires even though it is expensive. This is because copper is a good conductor of electricity.

Write down one other reason why copper is used to make electric wires. Ductile

b) The diagram below shows the structure of a metal.



Use ideas about the structure of metals to explain why

i) copper conducts electricity.
Copper has free electrons / has delocalized electrons.
ii) iron has a very high melting point.

(OCR GCSE 21st Century Science (Higher Tier), Additional Science B, Unit 1, B623/02, Jan. 2011, 6)

24 Copper is extracted from the ore chalcopyrite ($CuFeS_2$) in a three-stage process.

a) In the first stage of this extraction, the chalcopyrite is heated with silicon dioxide and oxygen.

Balance the following equation for this first stage in which copper(I) sulphide is formed.

$$\underline{2}\ CuFeS_2 + \underline{2}\ SiO_2 + \underline{4}\ O_2 \longrightarrow Cu_2S + \underline{2}\ FeSiO_3 + \underline{3}\ SO_2$$

b) In the second stage of this extraction, the copper(I) sulphide is converted into copper(II) oxide. This occurs by roasting the sulphide with oxygen at high temperature. Write an equation for this reaction.
$Cu_2S(s) + 2O_2(g) \longrightarrow 2CuO(s) + SO_2(g)$

c) In the third stage of this extraction, copper(II) oxide is reduced to copper by its reaction with carbon. Write an equation for this reaction.
$2CuO(s) + C(s) \longrightarrow 2Cu(s) + CO_2(g)$

d) Scrap iron can be used to extract copper from dilute aqueous solutions containing copper(II) ions.

i) Explain why this is a low-cost method of extracting copper.
Scrap iron is cheap. / Low energy requirement.
ii) Write an ionic equation for the reaction of iron with copper(II) ions in aqueous solution.
$Fe(s) + Cu^{2+}(aq) \longrightarrow Fe^{2+}(aq) + Cu(s)$

(AQA Advanced Subsidiary GCE, Chemistry, Unit 2, Jan. 2010, 8)

25 W, X, Y and Z are four different metals. The table below lists the results of three experiments carried out using the metals or their oxides.

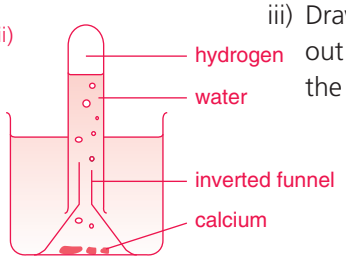
Experiment	W	X	Y	Z
Adding metal to cold water	no observable change	no observable change	formation of a colourless gas	no observable change
Adding metal to copper(II) sulphate solution	formation of a reddish brown solid	no observable change	formation of a colourless gas and a reddish brown solid	formation of a reddish brown solid
Heating the metal oxide with carbon	no observable change	formation of a solid with metallic lustre	no observable change	formation of a solid with metallic lustre

a) Arrange the metals in order of reactivity, starting with the most reactive one. Explain your answer briefly. Y > W > Z > X

b) Metal Y gives a brick-red flame in a flame test.

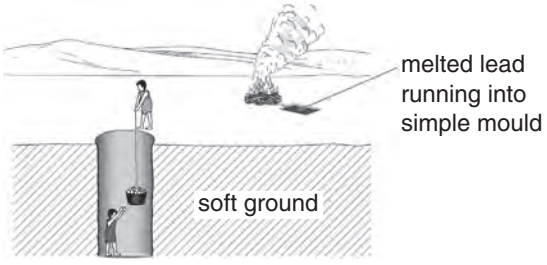
i) Suggest what metal Y might be.
Y might be calcium.
ii) Write a chemical equation for the reaction between Y and water.
$Y(s) + 2H_2O(l) \longrightarrow Y(OH)_2(s) + H_2(g)$
iii) Draw a labelled diagram of the set-up for carrying out the reaction between Y and cold water, with the collection of the gaseous product.

25 b) iii)



26 Read this article about lead mining.

> Archaeologists have discovered a small lead mine that was used 4 000 years ago.
>
> The miners made a deep hole through soft ground until they reached the lead ore. They then dug out the ore and loaded it into baskets.
>
> A second group of people lifted the ore out of the mine and put it into a wood fire. In the fire the ore reacted to make sulphur dioxide, carbon dioxide and melted lead. The melted lead was run into moulds and allowed to solidify.



a) Here is some information about three of the substances involved in the process.

	Lead	Lead ore	Sulphur dioxide
Melting point (°C)	327	1 114	–73
Boiling point (°C)	1 744	1 281	–10
Hazard	prolonged exposure to dust or fumes is harmful	prolonged exposure to dust or fumes is harmful	acidic gas

Use the information in the article and table above to suggest and describe at least FOUR likely risks to people. Explain how each risk is created and who would be affected.
Refer to page T54.
(Communication mark will be awarded for this part.)

b) Two reactions take place when lead ore is heated in the fire.

i) In reaction 1, lead(II) sulphide reacts with oxygen to make lead(II) oxide and sulphur dioxide. Balance the equation for this reaction.

$\underline{2}\ PbS + \underline{3}\ O_2 \longrightarrow \underline{2}\ PbO + \underline{2}\ SO_2$

ii) In reaction 2, the oxygen is removed from the lead(II) oxide.

What do we call a reaction in which oxygen is removed? Reduction

iii) What is the formula mass of lead(II) oxide, PbO? Formula mass of PbO = 207.2 + 16.0 = 223.2

(Relative atomic masses: O = 16.0, Pb = 207.2)

iv) Calculate the percentage by mass of Pb in the PbO. Refer to the next page.

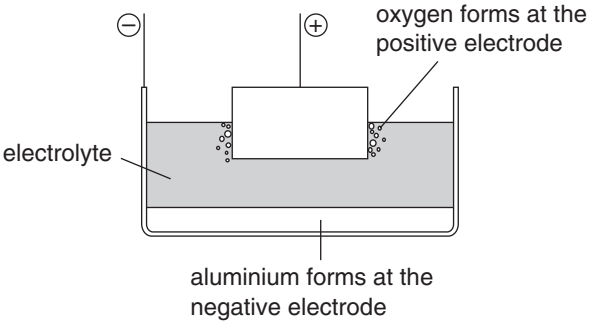
(OCR GCSE 21st Century Science (Higher Tier), Additional Science A, A152/02, Jun. 2012, 2)

27 a) Calculate the percentage of aluminium in aluminium oxide.

(Relative atomic masses: O = 16.0, Al = 27.0)

b) Aluminium is extracted from aluminium oxide using electrolysis.

The diagram shows a cell used for the extraction of aluminium.



a) Percentage by mass of Al in Al_2O_3

$= \frac{\text{number of atoms of Al in formula} \times \text{relative atomic mass of Al}}{\text{formula mass of } Al_2O_3} \times 100\%$

$= \frac{2 \times 27.0}{102.0} \times 100\%$

$= 52.9\%$

Topic Exercise

i) Cryolite lowers the melting point of aluminium oxide. Thus, less energy is needed to melt the oxide.

i) The electrolyte contains cryolite.

Explain why.

ii) Oxygen is formed at the positive electrode. Complete and balance the equation for this reaction.

$$\underline{2}\ O^{2-} \longrightarrow O_2 + \underline{4e^-}$$

iii) The positive electrode in the cell is used up during the process.

Explain why. The electrode is made from carbon / graphite. It reacts with oxygen to form carbon dioxide.

(AQA GCSE (Higher Tier), Chemistry, Unit 2, May 2012, 5)

28 Metals are usually extracted from oxides.

Some of these oxides occur naturally. Other oxides are made by roasting sulphide ores in air, producing sulphur dioxide as a by-product.

For the extraction of some metals, the oxide needs to be converted into a chloride.

a) i) The ore molybdenite contains molybdenum disulphide (MoS_2).

The first stage in the extraction of molybdenum is to roast the ore in air to form molybdenum oxide (MoO_3) and sulphur dioxide.

Write an equation for the first stage in this extraction.

$2MoS_2(s) + 7O_2(g) \longrightarrow 2MoO_3(s) + 4SO_2(g)$

ii) Pure molybdenum is formed in the second stage by the reduction of MoO_3 using hydrogen.

Write an equation for this reaction.

$MoO_3(s) + 3H_2(g) \longrightarrow Mo(s) + 3H_2O(l)$

iii) State ONE risk in using hydrogen gas in metal extractions.

Hydrogen is explosive / flammable / easily ignited.

b) Calcium is an expensive metal. It is extracted by the electrolysis of molten calcium chloride.

i) State why calcium chloride must be molten for electrolysis to occur. The ions become mobile when calcium chloride is molten.

ii) Identify the major cost in this extraction of calcium.

High electrical energy cost

(AQA Advanced Subsidiary GCE, Chemistry, Unit 2, June 2011, 4(a)–(b))

29 State and explain what you would observe when zinc granules are added to a solution of copper(II) sulphate. Write an equation for the reaction, including state symbols.

(Edexcel GCE O Level, Paper 1, Jun. 2010, 7(a))

30 Thermit reactions broadly refer to exothermic oxidation-reduction reactions between a metal powder and a metal oxide. One example is the reaction of finely divided iron(III) oxide with aluminium powder. This reaction results in a very high temperature, and is commonly used in the welding of rail tracks for trains. At this very high temperature, the molten iron formed joins the rail tracks together.

a) Complete and balance the chemical equation for the following thermit reaction.

$$___ Fe_2O_3(s) + Al(s) \longrightarrow _____ + _____$$

b) Copper powder CANNOT be used to replace aluminium powder in carrying out the thermit reaction with iron(III) oxide. Explain why.

c) The extraction of iron from its ores also involves the reduction of iron oxides.

i) Suggest why aluminium is NOT used as the reducing agent in iron extraction.

ii) Suggest ONE reducing agent commonly used in iron extraction.

(HKDSE, Paper 1B, 2013, 7)

Topic Exercise

29 A reddish brown coating is formed on zinc.
Copper is deposited on zinc.
Zinc displaces copper from copper(II) sulphate solution.
The blue colour of the solution fades / the solution turns colourless
because the concentration of copper(II) ions decreases.
$Zn(s) + CuSO_4(aq) \longrightarrow ZnSO_4(aq) + Cu(s)$

26 b) iv) Percentage by mass of Pb in PbO

$$= \frac{\text{number of atoms of Pb in formula x relative atomic mass of Pb}}{\text{formula mass of PbO}} \times 100\%$$

$$= \frac{207.2}{223.2} \times 100\%$$

$= 92.8\%$

31 Titanium is used in aircraft, ships and hip replacement joints. Titanium is as strong as steel but 45% lighter, and is more resistant to acids and alkalis.

Most titanium is produced from its ore, rutile (titanium oxide), by a process that takes up to 17 days.

Titanium oxide is reacted with chlorine to produce titanium chloride.

↓

Titanium chloride is reacted with magnesium at 900 °C in a sealed reactor for 3 days.

↓

The reactor is allowed to cool, then opened and the titanium is separated from the magnesium chloride by hand.

Titanium reactors produce about 1 tonne of the metal per day.

Iron blast furnaces produce about 20 000 tonnes of the metal per hour.

Refer to page T54.

a) Give ONE property of titanium that makes it more useful than steel for hip replacement joints.

b) In the reactor magnesium is used to produce titanium. If carbon were used instead of magnesium, no titanium would be produced.

What does this tell you about the relative reactivities of carbon, magnesium and titanium?

c) The use of titanium is limited because it is expensive.

Explain why titanium costs more than steel.

(AQA GCSE (Higher Tier), Chemistry, Unit 1, Jun. 2008, 4)

32 Ammonium phosphate is a fertilizer. The formula for ammonium phosphate is $(NH_4)_3PO_4$.



Refer to page T55.

a) Calculate the formula mass for ammonium phosphate.

149.0

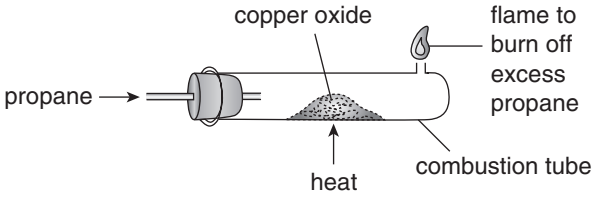
(Relative atomic masses: H = 1.0, N = 14.0, O = 16.0, P = 31.0)

b) Calculate the percentage by mass of nitrogen in ammonium phosphate. 28.2%

(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 2, Jan. 2012, 2(a))

33 A student wants to find the empirical formula of a sample of copper oxide.

The following is the apparatus she uses.



The student puts 2.86 g of copper oxide into the combustion tube. She passes propane gas over the heated copper oxide. After 20 minutes all the copper oxide has been changed into copper. She makes 2.54 g of copper in this experiment.

Refer to page T55.

a) i) What is the amount, in moles, of copper made?

0.0400 mol

(Relative atomic masses: Cu = 63.5)

ii) During the reaction all the oxygen in the 2.86 g of copper oxide is removed.

What is the amount, in moles, of oxygen atoms removed?

0.0200 mol

(Relative atomic masses: O = 16.0)

iii) Use your answers to parts (i) and (ii) to work out the empirical formula for copper oxide.

b) The student repeats the experiment.

This time she uses 5.72 g of copper oxide instead of 2.86 g.

5.08 g

What mass of copper should the student make?

(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 2, Jun. 2010, 6)

34 A compound of boron and chlorine has a relative molecular mass of 163.6 and contains 13.2% of boron by mass.

Calculate the molecular formula of this compound.
Refer to page T56.
(Relative atomic masses: B = 10.8, Cl = 35.5)

(AQA Advanced Subsidiary GCE, Unit 1, Jun. 2012, 5(e))

35 Hydrated calcium nitrate can be represented by the formula $Ca(NO_3)_2 \bullet xH_2O$ where x is an integer.

A 6.04 g sample of $Ca(NO_3)_2 \bullet xH_2O$ contains 1.84 g of water of crystallization.

Use this information to calculate a value for x.
4; refer to page T56.
(Relative atomic masses: H = 1.0, N = 14.0, O = 16.0, Ca = 40.1)

(AQA Advanced Subsidiary GCE, Unit 1, Jun. 2011, 2(c))

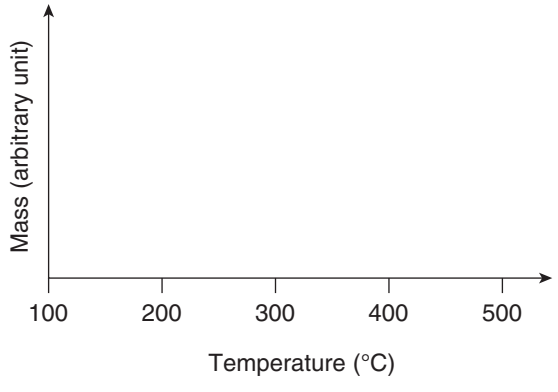
36 Copper(II) sulphate crystallizes from its aqueous solution as $CuSO_4 \bullet 5H_2O(s)$.

a) The water of crystallization of the salt can be liberated upon heating. Suggest a chemical test to show that water is being liberated.

b) Outline an experimental method to establish that the salt is a pentahydrate.

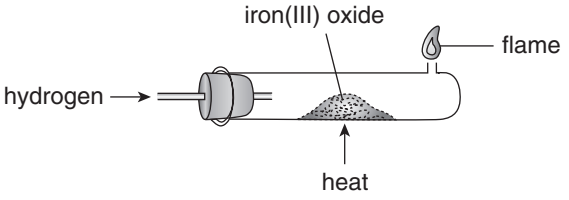
c) When $CuSO_4 \bullet 5H_2O(s)$ is heated slowly such that the temperature rises steadily, it will lose four water molecules at about 110 °C, and then the last water molecule at about 250 °C.

Using the axes below, sketch the change of mass when a sample of $CuSO_4 \bullet 5H_2O(s)$ is heated slowly.



(HKALE, 2011, Paper 1, 7(a))

37 The following experimental set-up was used to reduce 7.18 g iron(III) oxide to iron.



a) Write a chemical equation for the reaction that occurred in the experiment.

b) Suggest ONE way to show that a metal is formed in this experiment.

c) Calculate the mass of iron obtained.

(Relative atomic masses: O = 16.0, Fe = 55.8)

d) Explain why the excess hydrogen was burnt.

e) Draw a hazard warning label that should be displayed on a cylinder of hydrogen.

f) Upon heating, silver oxide decomposes to give silver and oxygen. Write a chemical equation for the decomposition.

g) Is it possible to deduce from the results of the experiments of iron and silver that iron occupies a higher position in the reactivity series than silver? Explain your answer.

37 a) $Fe_2O_3(s) + 3H_2(g) \longrightarrow 2Fe(s) + 3H_2O(l)$

b) The metal obtained can conduct electricity.

c) $Fe_2O_3(s) + 3H_2(g) \longrightarrow 2Fe(s) + 3H_2O(l)$
7.18 g ? g

Number of moles of Fe_2O_3 used $= \dfrac{\text{mass of } Fe_2O_3}{\text{molar mass of } Fe_2O_3}$

$= \dfrac{7.18 \text{ g}}{159.6 \text{ g mol}^{-1}}$

$= 0.0450 \text{ mol}$

According to the equation, 1 mole of Fe_2O_3 can produce 2 moles of Fe.

$\therefore$ number of moles of Fe obtained $= 2 \times 0.0450 \text{ mol}$
$= 0.0900 \text{ mol}$

Mass of Fe obtained
= number of moles of Fe x molar mass of Fe
$= 0.0900 \text{ mol} \times 55.8 \text{ g mol}^{-1}$
$= 5.02 \text{ g}$

d) Hydrogen gas is flammable and explosive. Burning the excess hydrogen avoids dangerous situation.

e)



f) $2Ag_2O(s) \longrightarrow 4Ag(s) + O_2(g)$

g) No. The reactivity of iron and silver can only be compared using the same reaction.

38 Copper(II) carbonate decomposes when heated. Copper(II) oxide and carbon dioxide are made.

$$CuCO_3(s) \longrightarrow CuO(s) + CO_2(g)$$

The following table shows the formula mass or relative molecular mass of each compound in the equation.

Compound	Formula mass / relative molecular mass
$CuCO_3$	123.5
CuO	79.5
CO_2	44.0

a) A student investigates the decomposition of copper(II) carbonate.

He predicts he should make 3.69 g of copper(II) oxide.

He actually makes 3.32 g of copper(II) oxide.

Calculate his percentage yield.

b) Another student heats 6.20 g of copper(II) carbonate.

Calculate the mass of copper(II) oxide that he should make.

(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 2, Jan. 2010, 3)

a) Percentage yield = $\frac{3.32\text{ g}}{3.69\text{ g}} \times 100\%$
= 90.0%

b) $CuCO_3(s) \longrightarrow CuO(s) + CO_2(g)$
6.20 g ? g

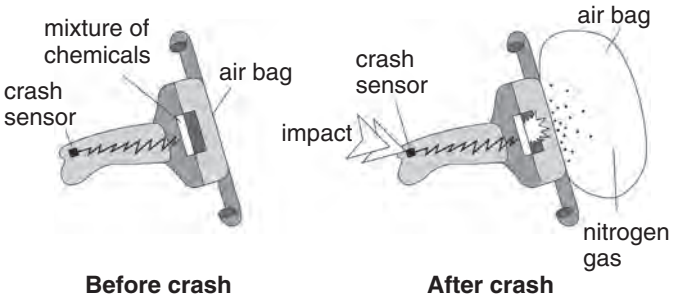
Number of moles of $CuCO_3$ = $\frac{\text{mass of } CuCO_3}{\text{molar mass of } CuCO_3}$
= $\frac{6.20\text{ g}}{123.5\text{ g mol}^{-1}}$
= 0.0502 mol

According to the equation, 1 mole of $CuCO_3$ can produce 1 mole of CuO.

∴ number of moles of CuO = 0.0502 mol

Mass of CuO = number of moles of CuO x molar mass of CuO
= 0.0502 mol x 79.5 g mol^{-1}
= 3.99 g

39 Air bags are used to protect the passengers in a car during an accident. When the crash sensor detects an impact, it causes a mixture of chemicals to be heated to a high temperature. Reactions then take place to produce nitrogen gas. The nitrogen fills the air bag.



a) The mixture of chemicals contains sodium azide (NaN_3) which decomposes on heating to form sodium and nitrogen.

$$2NaN_3(s) \longrightarrow 2Na(s) + 3N_2(g)$$

A typical air bag contains 130.0 g of sodium azide. Calculate the mass of nitrogen that would be produced when 130.0 g of sodium azide decompose.
84.0 g; refer to page T57.
(Relative atomic masses: N = 14.0, Na = 23.0)

b) The sodium produced when the sodium azide decomposes is dangerous. The mixture of chemicals contains potassium nitrate and silicon dioxide, both of which help to make the sodium safe.

i) Sodium reacts with potassium nitrate to make sodium oxide, potassium oxide and nitrogen. Complete the balancing of the equation for this reaction.

$$10Na(s) + \underline{2}\ KNO_3(s) \longrightarrow \underline{5}\ Na_2O(s) + K_2O(s) + N_2(g)$$

ii) The silicon dioxide reacts with the sodium oxide and potassium oxide to form silicates.

Suggest why sodium oxide and potassium oxide are dangerous in contact with the skin.

(AQA Advanced Subsidiary GCE, Unit 2, Jun. 2010, 4(a)–(b))

b) ii) Any one of the following:
- They are corrosive / burn.
- They are basic / alkaline.
- They attack living tissue.

Topic Exercise

40 Potassium superoxide is used to provide emergency supplies of oxygen in submarines.

The equation below shows the reaction of potassium superoxide that makes oxygen.

$$4KO_2(s) + 2CO_2(g) \longrightarrow 2K_2CO_3(s) + 3O_2(g)$$

Anthony reacts 71.1 g of potassium superoxide, KO_2, with excess carbon dioxide, CO_2. What mass of oxygen is made? 24.0 g; refer to page T57.

Use the relative molecular masses or formula masses in the table.

Substance	Relative molecular mass / formula mass
KO_2	71.1
CO_2	44.0
K_2CO_3	138.2
O_2	32.0

(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 2, Jun. 2011, 8(d))

41 Magnesium nitrate decomposes on heating to form magnesium oxide, nitrogen dioxide and oxygen as shown in the following equation.

$$2Mg(NO_3)_2(s) \longrightarrow 2MgO(s) + 4NO_2(g) + O_2(g)$$

Thermal decomposition of a sample of magnesium nitrate produced 0.741 g of magnesium oxide.

a) Calculate the amount, in moles, of MgO in 0.741 g of magnesium oxide.

b) Calculate the total amount, in moles, of gas produced from this sample of magnesium nitrate.

(Relative atomic masses: N = 14.0, O = 16.0, Mg = 24.3)

(AQA Advanced Subsidiary GCE, Unit 1, Jun. 2010, 3(a)(i)–(ii))

a) Number of moles of MgO $= \dfrac{\text{mass of MgO}}{\text{molar mass of MgO}}$

$= \dfrac{0.741\text{ g}}{40.3\text{ g mol}^{-1}}$

$= 0.0184$ mol

b) $2Mg(NO_3)_2(s) \longrightarrow \underset{0.0184\text{ mol}}{2MgO(s)} + \underbrace{4NO_2(g) + O_2(g)}_{?\text{ mol}}$

According to the equation, 2 moles of $Mg(NO_3)_2$ decompose to give 2 moles of MgO and 5 moles of gas.

$\therefore$ number of moles of gas $= \dfrac{5}{2} \times 0.0184$ mol

$= 0.0460$ mol

42 The ore haematite contains iron(III) oxide. Iron is extracted from this ore by reaction with carbon.

The products of this reaction are iron and carbon dioxide.

a) Finish this equation for the reaction.

$$\underline{2}\ Fe_2O_3(s) + \underline{3}\ C(s) \longrightarrow \underline{4Fe}\ (s) + \underline{3CO_2}\ (g)$$

b) A haematite ore contains 80.0% by mass of iron(III) oxide.

Calculate the maximum mass of iron that can be extracted from each tonne of this ore.

(1 tonne = 1 000 kg)
0.559 tonne; refer to page T58.
(Relative atomic masses: O = 16.0, Fe = 55.8)

(OCR GCSE 21st Century Science (Higher Tier), Chem. A, Unit 2, Jan. 2008, 5)

43 A metal carbonate decomposes according to the equation:

$$MCO_3(s) \longrightarrow MO(s) + CO_2(g)$$

When 5.02 g of MCO_3 are decomposed completely by heating, 1.76 g of carbon dioxide are produced. Calculate the molar mass of MCO_3 and hence identify M. Refer to page T58.

(Relative atomic masses: C = 12.0, O = 16.0)

44 A worker is fixing the roof of his shed using galvanized iron nails.



Galvanized iron nails have been coated in zinc.

The zinc coating prevents the iron nail from rusting.

Explain TWO ways the zinc coating does this.

(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 2, Jun. 2010, 10(a))

The zinc coating stops water and oxygen from reaching the iron.
Zinc acts as a sacrificial metal.

45 A lady has an old Landrover. Some parts of the Landrover is made from aluminium.



a) Explain why aluminium does not corrode.

b) The table below shows some information about iron and aluminium.

	Aluminium	**Iron**
Density (g cm^{-3})	2.7	7.9
Relative strength	1	2.7
Effect of moist air	no corrosion	rusts quickly

i) Write down ONE advantage of using aluminium to make car bodies.

ii) Write down ONE disadvantage of using aluminium to make car bodies.

c) European law says that 85% of a car must be able to be recycled.

Give a reason, other than cost, why it is good to recycle parts of a car.

(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 1, Jun. 2009, 7)

a) Aluminium reacts with oxygen in the air to form an even coating of oxide.
This oxide layer sticks firmly to the surface of the metal and stops oxygen / air / water from reaching the metal.
b) i) Low density / the car body is lighter.
ii) Not so strong
c) Any one of the following:
- Less metal ore needed
- Less waste

(Any other acceptable answer)